

SANDEEP KAHAWATHA

Software Engineer | Full-Stack Developer | AI/ML Researcher

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PROFESSIONAL SUMMARY

Results-driven Software Engineering undergraduate at SLIIT (Class of 2026) specializing in full-stack web and mobile development. Expert in the MERN stack, NestJS, TypeScript, and React Native with production-grade experience across web applications, LMS, mobile, e-commerce, and real-time dashboard platforms for clients across multiple domains. Published final-year researcher in Multi-Agent Reinforcement Learning and Graph Neural Networks, achieving an 81.8% reduction in simulated urban traffic waiting time. Passionate about clean architecture, scalable microservices, and delivering measurable real-world impact through software.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, PHP, C, C++, SQL

Frontend: React.js, Next.js, React Native, HTML5, CSS3, Tailwind CSS, Bootstrap, Material UI

Backend: Node.js, Express.js, NestJS, Flask, Laravel, SOLID Principles, Clean Code, RESTful API Design

Databases: MongoDB, MySQL, Firebase, Redis, Supabase

AI / ML: PyTorch, PyTorch Geometric, Graph Neural Networks (GNN), MAPPO, Reinforcement Learning, Bayesian UQ

DevOps & Tools: Git, Docker, Docker Compose, WebSockets, JWT, REST APIs, Agile/Scrum

Design & Other: Figma, Chart.js, Recharts, SUMO Simulator, Jest, Supertest

PROJECTS

Multi-Agent Urban Traffic Signal Optimization | Final Year Research Project

Python · PyTorch · PyTorch Geometric · GNN · MAPPO · Flask · React · Redis · Docker · SUMO | April 2026

- Designed UQ-RecurrentHGAT a novel AI architecture fusing Heterogeneous Graph Attention Networks (HGAT) with Multi-Agent Proximal Policy Optimization (MAPPO) for real-time adaptive signal control across entire city networks.
- Engineered a physically-grounded heterogeneous graph (Lane + Intersection nodes) with static edge attributes enabling lane-capacity-aware decisions; implemented 2-Hop Residual HGAT with skip connections to prevent over-smoothing and integrated a GRU temporal memory module.
- Integrated Bayesian Epistemic Uncertainty Quantification via vectorized Monte Carlo Dropout ($K = 20$ passes); achieved P50 inference latency of ~ 3.17 ms, validating edge-computing deployment viability.
- Benchmarked vs. SUMO actuated baselines key results:** 81.8% reduction in average network waiting time, 62.2% reduction in queue length, and 79.7% reduction in total weighted cost.

Learning Management System (LMS) | [Link](#)

Next.js · Supabase · WebSockets · Chart.js

- Architected a full-featured multi-role LMS from scratch supporting Admin, Teacher, and Student portals with JWT-based authentication and role-based access control.
- Implemented real-time batch chat (WebSockets / Supabase Realtime), a live bell-icon notification system, and a dual-mode anti-fraud QR attendance engine with rotating codes and camera scanning.
- Developed a secure timed Exam Engine supporting auto-graded MCQs, manual essay scoring, timer auto-submit, and downloadable PDF certificate generation.
- And analytics dashboards (Chart.js / Recharts) for attendance drop-off rates and grade bell-curve insights.

MiniMed - Mobile Health Application | [Repository1](#) | [Repository2](#)

React Native

- Developed an age- and medical-history-based Vaccination Management System with push notifications for upcoming doses, user confirmation flows, and allergy and health-condition management with clinical alerts.

Elber - E-Commerce Platform | Full-Stack Developer | [Link](#)

NestJS · Next.js · React · MongoDB · TypeScript · Firebase Storage

- Developed a production-ready e-commerce platform covering product catalog, shopping cart, and checkout workflows; built NestJS RESTful API with Mongoose ORM and Firebase Storage integration for media hosting.

Ceylon Herb Care - Herbal E-Commerce Platform | [Repository](#)

MERN Stack

- Built a Seller Partnership Management Module with onboarding, authentication, automated email notifications, and dynamic historical performance reporting.
- Developed a comprehensive Order Management System with real-time order-status notifications, multiple payment method support, incomplete-payment alerts, and automated invoice generation.

Personal Finance Tracker | Backend System | [Repository](#)

Node.js · Express · MongoDB · JWT · Jest

- Engineered a backend system for income, expense, budget, and goal management with JWT authentication, Currencylayer API multi-currency support, and a Jest / Supertest test suite ensuring production reliability.

Secure Multilingual E-Voting System | [Repository](#)

React · Node.js · Express · MongoDB · JWT

- Developed a Sinhala / Tamil / English e-voting platform with NIC-based authentication, one-vote-per-voter enforcement, and a stress-tested architecture validated for high-traffic resilience.

Techart Solutions - RFID Web Application | [Repository](#)

Laravel

- Built an enterprise RFID management system covering device configuration, tap processing, and full audit-history management with seamless database integration.

Fashion Globe - Sales Optimisation System | [Repository](#)

MERN Stack

- Built a Customer Feedback and Loyalty Management System with product reviews, points tracking, personalized reward catalog, seasonal discounts, and engagement analytics dashboards.

CleanCity - Smart Waste Collection System | [Repository](#)

React · Firebase

- Designed administrative features for staff management, task assignment, collection scheduling, and route optimisation; built reporting frameworks for operational data analysis and performance tracking.

EDUCATION

BSc (Hons) in Information Technology - Specialization in Software Engineering

Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka | 2022 – Expected Graduation: 2026

G.C.E. Advanced Level Examination - Physical Science Stream

St. Anne's Collage, Kurunegala | August 2020

- Results: 3C Passes

ACHIEVEMENTS & CERTIFICATIONS

- Educational Achievement Certificate - IDEALIZE 2024, AIESEC in University of Moratuwa (Open Category Teams)
- Python for Beginners Certificate - CODL, University of Moratuwa (Code: GLd2Hz0BCL)
- Python Programming E-Certificate - CODL, University of Moratuwa